Eight lectures on:

- LTI Systems (lecture 7 & 9)
- Z Transform (lecture 8)
- FIR Filters (" 8)
- IIR Filters (" 10)
- Implementation Aspects (lecture 11)
- CTFT, DTFT, DFT (lecture 12)
- FFT (lecture 13)

Lecture 14 → Exam Samples

Lecture 7 :

1 - Introduction

2 - Classification of Systems

3 - LTI System's Output

4 - Convolution

5 - Convolution's Properties

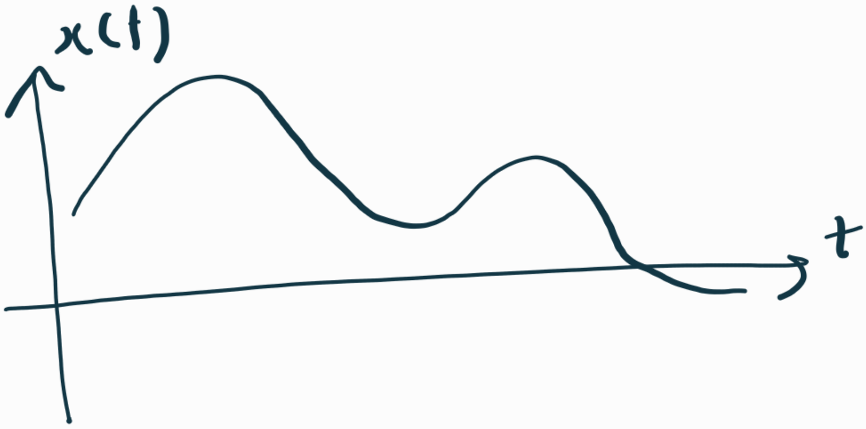
6 - Constant-Coefficient Difference Equations

1. Introduction:

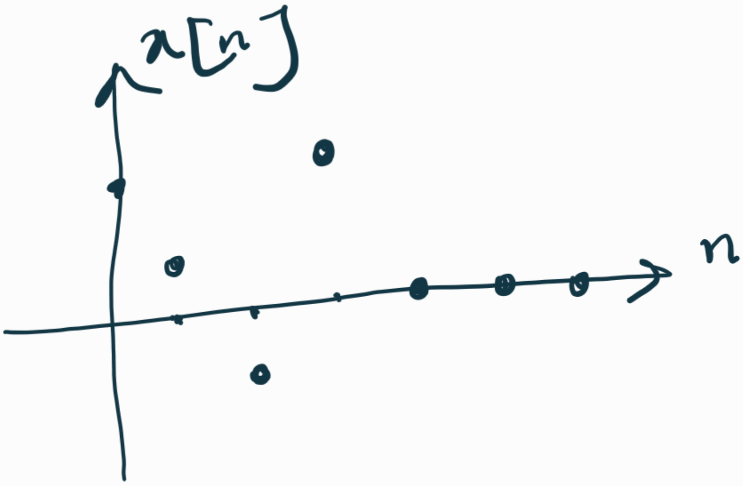
Signal : Any quatity that varies with time (or space, ...)

Examples: Audio, EEG, ECG, Radio waves, Images, videos, ...

- Continuous - Time Signal $\rightarrow x(t)$
  Is defined for every instant of time (time variable is continuous)



- Discrete - Time Signal $\rightarrow x[n]$
  Is defined only at specific time instants (time variable takes integer values)



System: Anything that takes an input signal and produces an output signal

$$x[n] \longrightarrow \boxed{\text{DT System}} \longrightarrow y[n]$$

$$x(t) \longrightarrow \boxed{\text{CT System}} \longrightarrow y(t)$$

Examples: Amplifier, Filter, DSP Algorithm

Communication Channel, ...

2. Classification of Systems:

- Static (without Memory) vs Dynamic

- Time-Invariant vs Time-Variant

A system that its input-output charactristics does not change with time

For CT Systems: if $x(t) \to y(t)$ then

$$x(t-t_0) \to y(t-t_0)$$

For DT Systems: if $x[n] \to y[n]$ then

$$x[n-n_0] \to y[n-n_0]$$

- Linear vs Non-linear

A system that satisfies the superposition principle:

For CT Systems:

$$x_1(t) \to y_1(t) \qquad \overset{\text{The}}{\Longrightarrow}$$
$$x_2(t) \to y_2(t)$$

$$a_1 x_1(t) + a_2 x_2(t) \to a_1 y_1(t) + a_2 y_2(t)$$

For DT Systems:

$$x_1[n] \to y_1[n] \qquad \overset{\text{Then}}{\Longrightarrow}$$
$$x_2[n] \to y_2[n]$$

$$a_1 x_1[n] + a_2 x_2[n] \to a_1 y_1[n] + a_2 y_2[n]$$

- Causal: The output only depends on current and past inputs (not future ones)

  past → can be dynamic $\qquad y(t) = f(x(t-k))$, $\underset{\sim}{k \geq 0}$

(It does not anticipate future inputs)
& we can have real-time processing

- Bounded Input - Bounded Output (BIBO) Stable

A system that produces bounded output ($|y[n]| \leq M_y$) for bounded inputs ($|x[n]| \leq M_x$)

↓

Why is it important?

As if a system is not BIBO stable, small disturbances, noises / inputs can lead to uncontrollable or unbounded outputs which can couse damage to electronic devices.

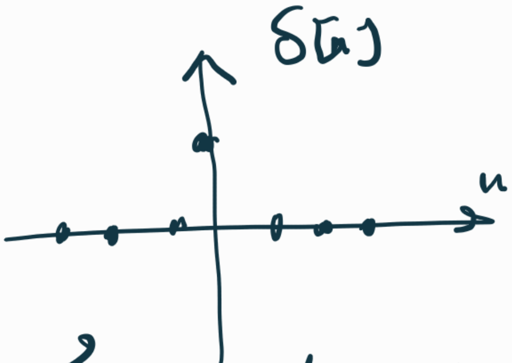
Our focus → $\boxed{\text{Linear Time-Invariant Systems (LTI Systems)}}$

3. LTI System's Output

Impulse Response, $h[n]$, of a system:

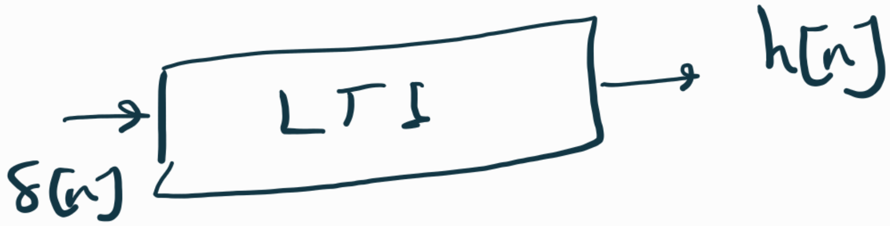
The output of the system when the input is $\delta[n]$

A unit impulse at $n=0$



$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & \text{otherwise} \end{cases}$$

Let's consider an LTI system



<u>DT:</u> Write $x[n] = \{\underline{1}, 2, 3\}$ as a sum of delayed dirac function ($\delta[n-k]$).

$$x[n] = \delta[n] + 2\delta[n-1] + 3\delta[n-2]$$

Hence, an arbitrary input $x[n]$ can be written as : $x[n] = \sum_{k=-\infty}^{+\infty} x[k]\,\delta[n-k]$

Now if $\delta[n] \to h[n]$ & system is LTI :

$$\delta[n-k] \xrightarrow[\text{-Invariant}]{\text{Time}} h[n-k]$$

Linearity $\longrightarrow$ $\underbrace{\sum_{k=-\infty}^{+\infty} x[k]\,\delta[n-k]}_{x[n]} \longrightarrow \underbrace{\sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]}_{\sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]}$

$$\underbrace{x[n]} \longrightarrow \underbrace{\sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]}$$

$$\triangleq \underbrace{x[n] * h[n]}$$

This is expression for DT convolution of $x[n]$ and $h[n]$

$$\boxed{x[n] \xrightarrow{LTI} x[n] * h[n]}$$

4. Convolution : Tabular Method

For $x[n] = \begin{cases} x_n & 0 \leqslant n \leqslant N-1 \\ 0 & \text{else} \end{cases}$

$h[n] = \begin{cases} h_n & 0 \leqslant n \leqslant M-1 \\ 0 & \text{else} \end{cases}$

$x[n] * h[n] \rightarrow$ has the length of $\boxed{\underbrace{N+M-1}}$

Tabular Method:

Step 1: Create the Conv. Table.

(a) The first row represents one of the sequences/signals ($x[n]$)

(b) The first column represents the other one ($h[n]$)

(c) multiply the first row by the enteries of the first column.

Step 2: Obtain $y[n]$ enteries by summing diagonals from bottom_left to top_right

(The length of $y[n]$ = length of $x[n]$ + Length of $h[n]$ − 1)

$\boxed{\text{Slide}}$

$h[n] = \{\underline{1}, 2, 3, 0, 1\}$ → length = 7

$x[n] = \{1, 1, 1\}$

DT2 → Example

5. Convolution's Properties

① Commutative law:

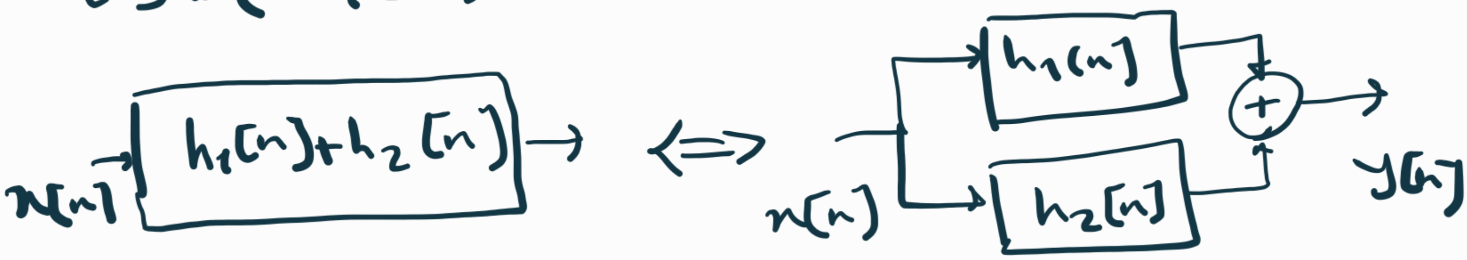
$$h[n] * x[n] = x[n] * h[n]$$

② Associative Law:

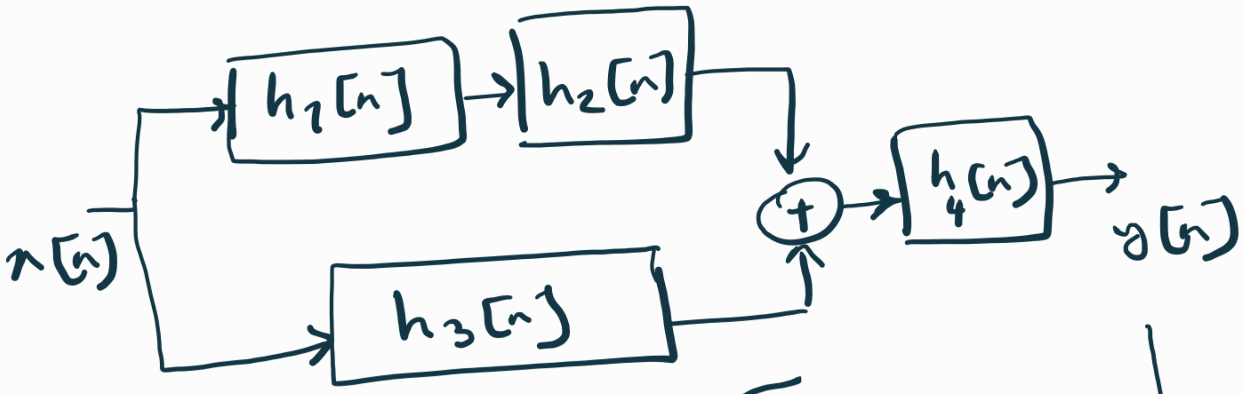
$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$

③ Distributive Law:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$



DT3: what is the impulse response of:



$$(h_1[n] * h_2[n]) + h_3[n]$$

$$h_{eq}[n] = \left[(h_1[n] * h_2[n]) + h_3[n]\right] * h_4[n]$$

6. Constant-Coefficient Difference Equations:

They are describing LTI systems:

$$y[n] = -\sum_{k=1}^{N} a_k y[n-k] + \sum_{k=0}^{M} b_k x[n-k]$$

We have 2 types of systems / filters:

① If $a_k = 0$ for all $k \rightarrow y[n] = \sum_{k=0}^{M} b_k x[n-k]$

It is an FIR (Finite Impulse Response) ~~system~~ filter

② Otherwise → It is an IIR (Infinite Impulse Response) System/filter

why are the Impulse Response in FIR filters finite?

$$y[n] = \sum_{k=0}^{M} b_k \, x[n-k] \longrightarrow$$

$x[n-k] = x[n] * \delta[n-k]$

$$= \sum_{k=0}^{M} b_k \, x[n] * \delta[n-k]$$

$$= x[n] * \underbrace{\left(\sum_{k=0}^{M} b_k \, \delta[n-k] \right)}_{h[n]}$$

$h[n]$ → The summation of M delayed scaled impulses.

$$\boxed{\begin{array}{l} \text{3 Important Properties of } \delta[n] \\ ①\ \sum_{n=-\infty}^{+\infty} \delta[n] = 1 \\ ②\ x[n] \cdot \delta[n-k] = x[k] \\ ③\ x[n] * \delta[n-k] = x[n-k] \end{array}}$$